ELSEVIER

Intern. J. of Research in Marketing 15 (1998) 37–50

International Journal of
Research in
Marketing

An investigation into the association pattern technique as a quantitative approach to measuring means-end chains

Frenkel ter Hofstede [a,b,*], Anke Audenaert [c], Jan-Benedict E.M. Steenkamp [a,d], Michel Wedel [b]

[a] *Department of Marketing and Marketing Research, Wageningen University, Wageningen, The Netherlands*
[b] *Department of Business Administration and Management Science, University of Groningen, Groningen, The Netherlands*
[c] *Unilever Research Laboratory, Colworth, UK*
[d] *Department of Marketing, Catholic University of Leuven, Leuven, Belgium*

Received 15 August 1996; accepted 17 July 1997

Abstract

Means-end chain theory links products to consumers by postulating hierarchical relations between attributes of the product, consequences of product use and values of consumers. It has served as an important conceptual framework for studies in marketing. The authors investigate the association pattern technique (APT) as a supplement to laddering, the most popular, qualitative measurement methodology in means-end chains research. APT is a structured method for measuring means-end chains, suitable for large-scale surveys. It can be used in personal as well as quantitative mail interviews. APT separately measures the attribute–consequence, and the consequence–value links. The independence of attribute–consequence, and consequence–value links is crucial to the validity of APT. Using loglinear models, we investigate this assumption for empirical data on four different products. Consistent support for independence is found. In addition, we use loglinear models to test the convergent validity of APT and laddering with respect to the content and structure of the means-end chains network that they reveal. The results show that the content of the APT and laddering networks differs. This result is explained from the different task formats. Most importantly, the hypothesis that the structure of APT and laddering networks is the same could not be rejected. © 1998 Elsevier Science B.V.

Keywords: Laddering; Measurement; Conditional independence; Convergent validity

1. Introduction

Means-end chain theory has proven to be useful in understanding consumer behavior (Peter and Olson, 1987; Valette-Florence and Rapacchi, 1991; Aurifeille and Valette-Florence, 1995; Pieters et al., 1995). Means-end chain theory builds upon work of psychologists (Tolman, 1932) and economists (Abbott, 1955) who have long recognized that consumers do not buy products for the product's sake, but for what the product can do for them. The theory relates the product to the consumer by positing a hierarchical cognitive structure involving linkages between attributes of the product, consequences of

* Corresponding author. Tel.: +31-317-483385; fax: +31-317-484361; e-mail: frenkel.terhofstede@alg.menm.wau.nl. This article was submitted, underwent blind review, and was accepted for publication, during the editorship of John Saunders.

0167-8116/98/$19.00 © 1998 Elsevier Science B.V. All rights reserved.
PII S0167-8116(97)00029-3

product use, and values of consumers. These concepts constitute the *content* of consumer knowledge, whereas the linkages between the concepts form the *structure*. Means-end chain theory, thus, is concerned with the content and the structure of consumer knowledge. It extends previous work in psychology and economics in that it is more specific in what the content and structure of consumer knowledge is, and how it can be assessed and applied to practical marketing problems, such as advertising (Reynolds and Craddock, 1988) and product development (Reynolds and Gutman, 1988).

In means-end chain theory, products are seen as means through which consumers obtain valued ends. According to this theory, consumers choose products because they believe that the specific attributes of the product can help them to achieve desired values through the consequences or benefits of product-use (Reynolds and Gutman, 1984). Attributes are the concrete, tangible characteristics of the product. Consequences refer to what the product does or provides to the consumer at the functional or psychosocial level. Values are intangible, higher-order outcomes or ends, being cognitive representations of consumers' most basic and fundamental needs and goals. The three levels: attributes, consequences and values are postulated to be hierarchically structured in that attributes lead to consequences, which produce value satisfaction. Although most models of cognitive structure specify some kind of hierarchical structure (e.g. Bettman, 1974; Schank, 1982), the defining feature of means-end chain theory is the specification of the exact linkages between the levels in the hierarchy, constituting the structure of consumer knowledge.

Despite the attention means-end chain theory has generated, as evidenced by the 1995 special issue of the International Journal of Research in Marketing, among others, there are still a number of issues that need to be addressed (cf. Grunert and Grunert, 1995). One particularly important issue concerns the methodology to measure consumers' means-end chains. Laddering is by far the most popular methodology (Reynolds and Gutman, 1988; Claeys et al., 1995). It is a qualitative interviewing technique and has been applied successfully in academic and applied research, but it is not without its limitations. Laddering interviews are time-consuming, costly, and require highly trained interviewers. Consequently, it is difficult, if not impossible, to use them to obtain the large-scale representative samples that are typically required for market forecasts and segmentation, among others.

In this paper we describe and investigate a quantitative methodology to measure means-end chains, which we will denote as the association pattern technique (APT). APT uses a fixed-format, and measures the links between attributes and consequences, and the links between consequences and values separately. The technique is much cheaper and faster than laddering, can be used in mail questionnaires, and allows the researcher to collect data among a representative sample of consumers. Thus, APT may be seen as an attractive option to measure means-end chains when large samples are needed. The purpose of this paper is to empirically explore the validity of APT. More specifically, we investigate the crucial assumption made in APT, viz. the independence of the attribute–consequence (AC) linkages and consequence–value (CV) linkages. In addition, the convergent validity between APT and laddering in terms of content and structure of the means-end chains is investigated.

2. Techniques for measuring means-end chains

2.1. Laddering

In the literature several approaches have been suggested to reveal means-end structures prevalent among consumers: laddering is the most widely applied technique (Reynolds and Gutman, 1988). In fact, sometimes laddering is equated with means-end chain theory, although the theory should be considered separate from the methodology. In the laddering procedure, three steps are distinguished: (1) elicitation of salient attributes, (2) the depth-interview, and (3) analysis of the results. In the first phase, the attribute-elicitation phase, consumers are questioned about the attributes used to compare and evaluate products. Several techniques may be used to elicit the important attributes, including the repertory grid, stimulus grouping, and direct elicitation.

In the second step, the most important attributes identified in the first phase are used as a starting

point for the depth-interview. The consumer is continuously probed with some form of the question 'why is that important to you?'. This way of questioning forces the subject up the 'ladder' of abstraction, until s/he cannot go further. The end need not always be at the value level. The result is a sequence of concepts, which are called ladders.

In the third step, the idiosyncratic concepts resulting from the laddering interviews are categorized into a smaller number of categories. The linkages between the (categorized) concepts are represented in the so-called implication matrix. This is an asymmetrical dominance matrix in which the concepts constitute both the rows and the columns. The cell entries indicate the frequencies, across all subjects in the sample, with which an attribute, consequence or value (the row element) leads directly or indirectly (through one or more other concepts) to another attribute, consequence or value (the column element). The implication matrix preserves information about the sequence of concepts in the means-end structure, but discards differences between ladders of the same or different individuals. From the implication matrix, the so-called hierarchical value map is constructed, depicting the content and structure of consumer knowledge in a graphical way. The map gives an aggregate network representation of the means-end chains for the product in question (Reynolds and Gutman, 1988).

In the classical laddering procedure described above, the natural flow of speech of the consumer is restricted as little as possible. This kind of laddering is referred to as 'soft laddering'. On the other hand, some researchers used the 'hard laddering' approach, allowing less freedom in the answers of consumers and forcing consumers to follow one ladder at a time, in which each subsequent answer is on a higher level of abstraction (Grunert and Grunert, 1995). Examples of this 'hard laddering' approach are the self-administered paper and pencil method (Walker and Olson, 1991; Young and Feigin, 1975) and computerized data collection methods.

Laddering has served as a very useful qualitative technique to reveal means-end chains. However, it also has its limitations. Because the laddering interview is time consuming and must be carried out by trained interviewers, it is an expensive data collection technique. Moreover, it places a serious burden on respondents, and the quality of the data may be affected by respondent fatigue and boredom (cf. Steenkamp and Van Trijp, 1996). In sum, laddering is not suitable as an instrument to be used in large representative samples, nor was it intended to be used in this context.

In reaction to these limitations there have been several attempts to quantify means-end chains in large-scale studies using other methods. Vanden Abeele (1992) used a technique where respondents evaluate complete ladders. These ladders are explicitly described in laddering statements and rated on their credibility. The method may be used in questionnaires for large studies and provides quantitative information on ladders. However, the ladders are defined a priori and the number of ladders that may be processed is limited. Grunert (1997) used an extended form of conjoint analysis to quantify means-end chains of quality perceptions. Respondents rate a few product profiles on several characteristics. Structural equations models were used to estimate relations between characteristics at the aggregate level. Still the question remains as to how the estimates relate to the actual strengths of linkages. In this paper we propose the association pattern technique (APT) as a quantitatively oriented technique to assess means-end chains.

2.2. *The association pattern technique*

The association pattern technique is inspired by Gutman (1982). Gutman proposed that, for measurement purposes, the means-end chain can be conceived as a series of connected matrices. In APT an *AC*-matrix (attribute–consequence matrix) and a *CV*-matrix (consequence–value matrix) are distinguished. Fig. 1 gives an example of *AC*- and *CV*-matrices that were used in our empirical study (see below). In the *AC*-matrix, the a priori defined attributes and consequences are listed in the columns and rows, respectively, resulting in a table of all combinations of attributes and consequences. Similarly, the *CV*-matrix includes all possible combinations of the consequences and values. For each column in the *AC*-matrix (*CV*-matrix), respondents indicate to which consequences (values) that attribute (consequence) is perceived to lead. This results in a dataset of binary observations. Note that, in contrast

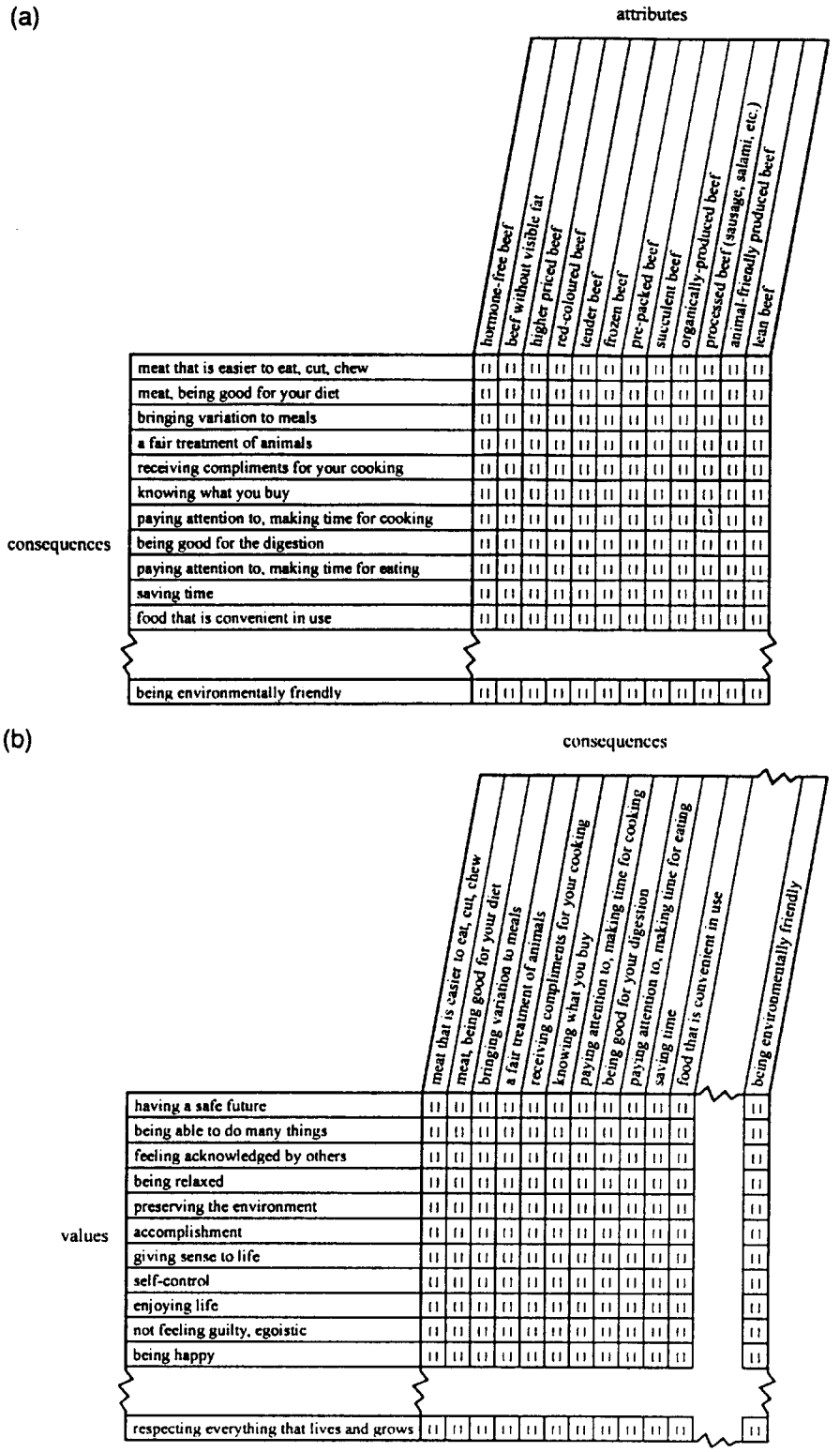


Fig. 1. Example of the association pattern technique. (a) The APT *AC*-matrix for beef, (b) the APT *CV*-matrix for b

to laddering, the attributes, consequences and values are to be provided by the researcher. Since the provided concepts need to be relevant and need to cover the range of concepts that constitute the content of means-end chains, pretesting is inevitable when secondary sources are lacking.

Given the limitations of laddering, the association pattern technique may serve as a useful supplement to laddering for measuring means-end chains. APT addresses the above mentioned limitations of laddering. The data-collection process is structured and it can be used in large-scale studies with appropriate control over representativeness of the sample on important characteristics. Obvious advantages accrue, for example in international marketing research. Furthermore, the analysis of the data is simple, especially because of the standardization of the concepts used in the matrices. This renders content analysis unnecessary.

3. Key research issues

Given these advantages of APT, the legitimacy of APT as a method to uncover means-end chains becomes an important issue. We make a distinction between two key research issues: the assumption of independence of the *AC*- and *CV*-linkages, and the convergent validity between APT and laddering. Below, we will elaborate further on these two issues.

3.1. Assumption of conditional independence

The basic assumption underlying APT is that *AC*- and *CV*-linkages are independent. This is a consequence of the separation of means-end chains in *AC*- and *CV*-matrices. The APT procedure implicitly assumes that the link of a consequence to a certain value in the *CV*-matrix is independent of the link a person previously indicated between an attribute and that particular consequence in the *AC*-matrix. In fact, stating it in probabilistic terms, this specifically corresponds to the assumption that attributes and values are conditionally independent, given the consequences. If this assumption holds, it allows the researcher to collect information on *AC*- and *CV*-linkages in separate matrices, which is the cornerstone of the APT method and its associated advantages. If this assumption of conditional independence is incorrect, the validity of APT as a technique to uncover means-end chains is in doubt.

In order to elucidate the issue of conditional (in)dependence, consider the example represented in Fig. 2. It may represent part of a consumer's cognitive structure. Two ladders are distinguished, which share a common consequence ('healthy'). Whereas in the laddering interview all information is retained on the ladders 'lean beef' – 'healthy' – 'enjoying life' and 'hormone–free beef' – 'healthy' – 'safety', APT breaks the ladders in two parts: on the one hand the *AC*-linkages 'lean beef' – 'healthy' and

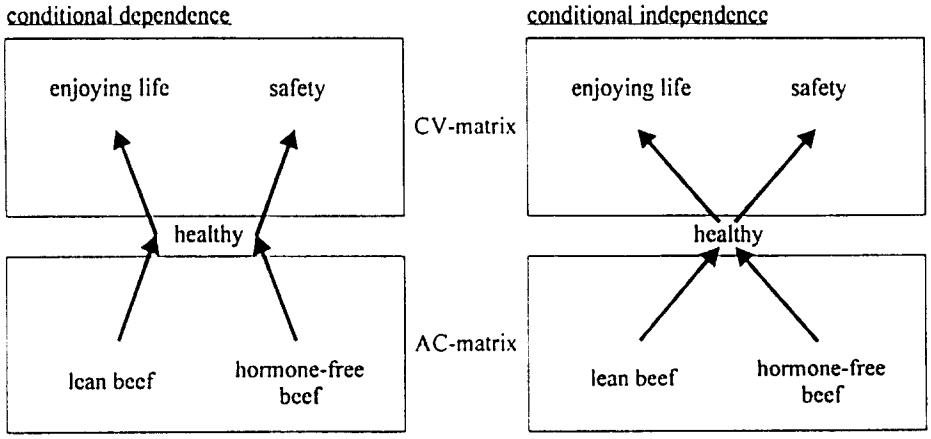


Fig. 2. Illustration of conditional dependence and independence of attributes and values for a given consequence.

'hormone-free beef' – 'healthy', and on the other hand the two *CV*-linkages 'healthy' – 'enjoying life' and 'healthy' – 'safety' and these *AC*- and *CV*-linkages are treated as being independent of each other. The question however, is whether in reality, given a consequence, specific attributes have a high probability of leading to specific values, as shown in the left panel of Fig. 2. If this is not the case, as shown in the right panel of Fig. 2, attributes and values are conditionally independent given the consequences.

Means-end chain theory is not clear as to whether conditional independence is to be expected or not. This is probably due to the dominance of laddering as a data collection technique in means-end chain research. Laddering accommodates both conditional dependence and independence situations, so this assumption did not need to be considered before. However, Gutman (1982) already proposed separating *AC*- and *CV*-linkages. Walker and Olson (1991) also view the consequences as intermediating concepts, separating the product (the attributes) from the self (the values) without suggesting any dependence in the linkages, while Peter and Olson (1987) note that numerous combinations of attributes, consequences, and values are possible. In sum, we expect conditional independence to be supported.

3.2. Between-method convergent validity

Between-method convergent validity is an important aspect for any set of techniques, including methods to uncover means-end chains. In this paper, we assess the convergent validity between APT and the 'standard' means-end chain measurement technique, laddering. Convergent validity is supported if cognitive structures identified with APT are similar to those obtained with laddering. We assess the similarity of APT and laddering with respect to the content as well as the structure of the means-end chains network.

The *content* of the means-end chains network refers to the nodes in the network. In this study, content is operationalized as the relative frequency at which the various attributes, consequences, and values occur. Complete similarity in content is attained if the concepts resulting from the laddering interviews and the responses in APT surface in equal proportions [1].

We define the *structure* of the means-end chains network as the relative frequency of occurrences of the linkages (irrespective of the proportions in which the specific attributes, consequences, and values occur). Structure refers to the strength of the linkages between the concepts. If similarity in content is lacking, this does not necessarily imply a dissimilarity in structure, or vice versa. Thus, content and structure are two different aspects of convergent validity assessment.

If certain differences in content and structure of cognitive structures measured by APT and laddering arise, they may be the result of at least two sources: (1) the response mode chosen, and (2) the large and a priori fixed set of attributes in APT versus the small and variable set of attributes in which each laddering interview starts. Below we will elaborate on these two differences.

In at least two ways, the response mode chosen may cause differences between APT and laddering in the means-end chains identified. First, the fixed-format of APT reduces the natural flow of speech, which increases the extent of strategic processing by the respondent (Grunert and Grunert, 1995). Strategic processes are those unwanted cognitive processes that do not result in activation of linkages between actual connected concepts, but occur when respondents choose alternative strategies just to arrive at an answer. On the other hand, the free-response format of laddering suffers from unwanted effects of interviewer involvement (Grunert and Grunert, 1995). Second, the free-response format in laddering implies that respondents have to recall the concepts from memory, whereas the fixed-format of APT is associated with recognition of these concepts that are presented to the subjects. This may lead to differences in the content and structure of the means-end chains activated, as recall and recognition are quite different processes (e.g. Alba and Hutchinson, 1987).

[1] Note that this definition also covers the situation where partially different sets of concepts occur in different methodologies, since then the frequency for a concept in one of the methods may equal zero.

The total number of recognized concepts typically exceeds the total number of recalled concepts (Singh and Rothschild, 1983). This suggests that the total number of attributes, consequences, and values is greater in APT than in laddering. In principle this need not hamper the comparability of the results of APT and laddering, provided the impact does not vary over the set of attributes, consequences, and values or, more importantly, the linkages between them. If however, the impact differs, both the content and structure of the means-end chains identified with laddering and APT may diverge.

A second source of differences between laddering and APT data may be the set of attributes used as a basis for data collection. In APT, all respondents are confronted with the same set of attributes, which may consist of more than 10 items. In the laddering interview only a small number of attributes can be used (usually three to five), which can differ between respondents. Only these attributes and attributes elicited through (indirect) linkages with these attributes appear in the ladders. Since attributes are linked to consequences, which are again connected to certain values, the content of the more abstract levels may be influenced, while the structure may also be affected.

In sum, we identify at least two reasons why APT and laddering do not necessarily yield similar results. Clearly, for the progress in the area of means-end chains, it is important to know to what extent the two techniques yield comparable results. Additionally and equally important, it needs to be investigated whether APT is fundamentally valid, in that the attributes and values are conditionally independent, given the consequences. In the remainder of this paper, we will empirically investigate the two key validity issues of conditional independence and convergent validity.

4. Method

4.1. Data

Data are obtained from a laddering study on food products in Belgium. Three hundred laddering interviews are conducted, concerning yogurt ($n = 100$), beef ($n = 100$), olive oil ($n = 50$) and vegetable oil ($n = 50$). The interviews are equally divided over Flanders and Wallonia, which are the Dutch- and French-speaking parts of Belgium, respectively. All respondents bought the product in question and the laddering interviews were conducted by trained interviewers. In this laddering study we slightly deviated from the classical laddering procedure in that attribute elicitation involved different respondents than the laddering interview itself.

In order not to overlook any key characteristic of a product, a prespecified list of attributes was constructed by pretesting in a separate elicitation phase on 20 respondents for each product. Three approaches were jointly used to develop sets of salient product attributes: a thorough review of the literature, extensive questioning on the product, and a repertory grid task (Kelly, 1955). The result of this final phase was a list of attributes for each product, which was used as point of departure in the depth-interview. Consumers were confronted with this list and were asked to indicate the importance of each attribute on a 3-point scale (not important to very important). This scale was used successfully in previous laddering research (Vanden Abeele, 1992). Three very important attributes and two rather important attributes were chosen by the interviewer as starting point for the laddering procedure. The raw data were content analyzed and categorized, and subsequently the implication matrix was constructed.

In addition, APT-data for beef were collected. The attributes included in the *AC*-matrix consisted of the list of the twelve most important attributes for beef used in the laddering study. The consequences and values included in the *AC*- and *CV*-matrices were selected on the basis of the frequency of their appearance in the implication matrix of the laddering study for beef. One hundred beef consumers completed the APT questionnaire, under the supervision of an interviewer. The *AC*- and *CV*-matrices included in the questionnaires are partially depicted in Fig. 1.

4.2. Analysis

We tested the conditional independence assumption and the between-method convergent validity, using a loglinear model formulation. The loglinear model describes the probabilities that the concepts and linkages among them occur in ladders. Modeling

the occurrence of ladders and linkages in a probabilistic framework is consistent with the ideas of Gutman (1991), who states that linkages "can be thought of in terms of some probability statement that one element will cause the occurrence of the other element (p. 144)."

4.2.1. Conditional independence

In order to test the assumption of conditional independence, we formulate a loglinear model representing the probability that a certain ladder occurs within a laddering interview. We start from the laddering data to test the assumption since it cannot by definition be tested on the APT data. From these data we form the three-way contingency table, indexed by A, C and V, that represents all ladders. In Eq. (1), the probability p_{ijk} that a ladder consists of attribute i, consequence j, and value k is expressed (on a logarithmic scale) as a linear function in the parameters. The set of parameters to be estimated consists of a scaling constant (α), the main effects, which capture the frequency of occurrence of attributes (β_i^A), consequences (β_j^C), and values (β_k^V), and their interactions (γ_{ij}^{AC}, γ_{jk}^{CV}, γ_{ik}^{AV}). The interactions capture the dependencies between attributes and consequences, consequences and values, and attributes and values respectively, i.e., they represent the direct and indirect linkages between the concepts. The saturated model is formulated as follows:

$$\ln p_{ijk} = \alpha + \beta_i^A + \beta_j^C + \beta_k^V + \gamma_{ij}^{AC} + \gamma_{jk}^{CV} + \gamma_{ik}^{AV} + \delta_{ijk}^{ACV} \quad (1)$$

where δ_{ijk}^{ACV} is the error term. The product attributes (i) range from 1 to I, consequences (j) range from 1 to J, and values (k) range from 1 to K. In loglinear modeling terms, we write this model as $[ACV]$ [2].

Within this framework we can test for the conditional independence (Agresti, 1990; DeSarbo and Hildebrand, 1980) of attributes and values, given consequences. In APT, the linkages A_iC_j and C_jV_k are separated. Therefore, as discussed in the previous section, APT assumes that these linkages are independent. Thus, if we know that a specific consequence is linked to a particular attribute, this should not provide any information about the value to which this consequence leads. In a probabilistic framework this comes down to the assertion that the probability that a particular C_jV_k-linkage is chosen (p_{jk}), does not depend on a preceding A_iC_j-linkage. In terms of probabilities of links, conditional independence implies $p_{ijk} = (p_{ij} \cdot p_{jk})/p_j$ (Agresti, 1990), which corresponds to the loglinear model $[AC, CV]$. In order to test whether this model adequately represents the laddering data, we test it against the model that does specify the direct dependence of attributes and values in addition to their indirect dependence through the consequences. This latter model includes the interaction term AV: $[AC, CV, AV]$. The former model is nested in the latter model. If AV-dependence exists, it will appear in the laddering interview, since the interview does not impose independence, and information is retained on individual ladders. Therefore we use laddering data to test for the conditional independence of the attributes and values given the consequences.

The 'fit' of loglinear models can be evaluated by means of likelihood-ratio test statistics (χ^2), indicating the fit relative to the saturated model $[ACV]$. This statistic follows a chi-square distribution with degrees of freedom equal to the number of cells in the table minus the number of linearly independent parameters. In addition, model fit will be evaluated using the normed fit index (Δ) of Bonett and Bentler (1983) which is independent of sample size. The index reflects the degree to which a model represents an improvement in goodness-of-fit (measured in χ^2) over the completely restricted model. When the model fully accounts for the data Δ takes on a value of 1.

Testing the conditional independence of the AC- and CV-linkages comes down to testing the hypothesis of no attribute-value interactions ($\gamma_{ik}^{AV} = 0$). The corresponding likelihood-ratio test statistic is the chi-square difference test statistic $\Delta\chi^2$: $\chi^2_{AC,CV} - \chi^2_{AC,CV,AV}$ (the difference in χ^2 between model $[AC, CV]$ and model $[AC, CV, AV]$). The statistic is chi-square distributed with $(I-1)(K-1)$ degrees of freedom under the null hypothesis of conditional

[2] This means that whenever the loglinear model contains higher-order effects, it also incorporates lower-order effects composed from the variables. For example, the model $[AC, V]$ corresponds to the loglinear model with intercept, three main effects A, C, V, and the first order interaction between A and C.

independence. If the test is not significant, the attributes and values are conditionally independent, which supports the procedure employed in APT of measuring *AC*- and *CV*-links in separate matrices.

4.2.2. Between-method convergent validity

The research question concerning the convergent validity between APT and laddering with respect to the content and structure of the means-end chains network, may also be formulated in terms of loglinear models. We collected both APT and laddering data for beef. We pool these data in order to test convergent validity, and form two 3-way contingency tables, one for the *AC*- and one for the *CV*-linkages. We introduce a new factor T in the loglinear models, representing the measurement technique from which the ladder originates: laddering ($t = 1$) or APT ($t = 2$). If the assumption of the conditional independence of attributes and values holds, we may treat the ladders as a sequence of independent linkages and formulate the (marginal) probabilities that linkages occur separately. In loglinear modeling terms we define the probabilities of occurrence of linkages as:

$$\ln p_{ijt}^{ACT} = \alpha + \beta_i^A + \beta_j^C + \beta_t^T + \gamma_{ij}^{AC} + \gamma_{it}^{AT} + \gamma_{jt}^{CT} + \delta_{ijt}^{ACT} \quad (2)$$

$$\ln p_{jkt}^{CVT} = \alpha' + \beta_j^{'C} + \beta_k^V + \beta_t^{'T} + \gamma_{jk}^{CV} + \gamma_{jt}^{'CT} + \gamma_{kt}^{VT} + \delta_{jkt}^{CVT} \quad (3)$$

Eqs. (2) and (3) are both saturated models. [3] The probability that an *AC*- or *CV*- linkage appears in the data collected with measurement technique t is reflected by p_{ijt}^{ACT} and p_{jkt}^{CVT}, respectively. Whereas α, α', β_i^A, β_j^C, $\beta_j^{'C}$, β_k^V, γ_{ij}^{AC}, and γ_{jk}^{CV} have the same interpretations as in Eq. (1), β_t^T and $\beta_t^{'T}$ capture the difference in the overall frequency of occurrence of concepts between the two measurement techniques, i.e., attributes and consequences taken together in Eq. (2), and both consequences and values in Eq. (3). The between-method difference in the frequency of occurrence of a specific attribute A_i, consequence C_j, or value V_k is taken into account by γ_{it}^{AT}, γ_{jt}^{CT}, $\gamma_{jt}^{'CT}$ and γ_{kt}^{VT}, respectively. In Eq. (2) the between-method difference in the frequency of occurrence, or strength, of a specific A_iC_j-linkage is captured by δ_{ijt}^{ACT}. Likewise, in Eq. (3), δ_{jkt}^{CVT} reflects between-method differences for the C_jV_k-linkages. According to our definition of content, the terms γ_{it}^{AT}, γ_{jt}^{CT}, $\gamma_{jt}^{'CT}$, and γ_{kt}^{VT} represent the differences in content of the cognitive network, whereas δ_{ijt}^{ACT} and δ_{jkt}^{CVT} reflect the differences in structure.

The test of between-method similarity of the content of the means-end chains network may be expressed as $\{\gamma_{it}^{AT} = \gamma_{jt}^{CT} = 0\}$ for the *AC*-part, and $\{\gamma_{jt}^{'CT} = \gamma_{kt}^{VT} = 0\}$ for the *CV*-part. This comes down to pairwise testing of several restricted models. For the *AC*-part of the data (Eq. (2)), we test for $\gamma_{it}^{AT} = 0$ and $\gamma_{jt}^{CT} = 0$, respectively, by testing the model $[A, C, T]$ against $[C, AT]$ and $[A, CT]$, all nested within Eq. (2). Thus, we test the model which assumes no between-method differences in the relative frequencies of the various concepts versus the models which account for these differences. Additionally we may test for $\gamma_{it}^{AT} = 0$ and $\gamma_{jt}^{CT} = 0$ simultaneously by testing the model $[A, C, T]$ against $[AT, CT]$. Likewise, in Eq. (3), these three tests come down to testing $[C, V, T]$ against the models $[V, CT]$, $[C, VT]$ and $[CT, VT]$. Analogous to the test of conditional independence, tests are based on the chi-square difference test statistic $\Delta\chi^2$, which is calculated from the difference in fit (χ^2) of each pair of models.

The similarity of structure may be assessed by testing $\{\delta_{ijt}^{ACT} = 0\}$ and $\{\delta_{jkt}^{CVT} = 0\}$. This corresponds to testing two restricted models from Eqs. (2) and (3), excluding the three-factor interaction terms δ_{ijt}^{ACT} and δ_{jkt}^{CVT}. Since the tests imply testing $[AC, AT, CT]$ and $[CV, CT, VT]$ against their saturated alternatives $[ACT]$ and $[CVT]$, the chi-square difference test statistic $\Delta\chi^2$ is identical to the χ^2 fit statistic of both models. Note here that by incorporating γ_{it}^{AT}, γ_{jt}^{CT}, $\gamma_{jt}^{'CT}$, and γ_{kt}^{VT} in Eqs. (2) and (3), we correct for the frequencies of the specific concepts in each measurement technique. If the chi-square difference statistic is significant for either the *AC*- or *CV*-part or both, we may conclude that APT and laddering do not reveal similar means-end chains in terms of structure.

[3] A prime is added to the terms in Eq. (3) that are also included in Eq. (2) since the parameters need not have the same values.

4.2.3. Cell-sparseness

The likelihood-ratio test statistic used to assess conditional independence and convergent validity is sensitive to cell-sparseness (Margolin and Light, 1974). Larntz (1978) indicated that the sample size should be at least five times the number of cells in the table. This condition may be violated in the context of laddering since many linkages may not occur in the data, resulting in zero-entries in the contingency tables. Such zero-entries should be considered as sampling zeros and not as structural zeros, since there are typically no theoretical arguments supporting that such links cannot occur. Violation of the Larntz criterion causes the chi-square tests for nested models to be over-conservative. Therefore, in the cases where the Larntz criterion is not satisfied, we adjust the chi-square and chi-square difference test statistics. As suggested by Fienberg (1980) we use Williams' (1976) formula, which comes down to dividing the chi-square test statistics by a correction factor. See Williams (1976) for mathematical details.

5. Results

5.1. Conditional independence

Since we are interested in testing the conditional independence of attributes and values, we focus on that part of the data where attributes and values come together at the intermediate level of the consequences. From the actual laddering interviews, we retain those ladders that contain complete attribute–consequence–value linkages, i.e. where an attribute is directly followed by a consequence and where this consequence is directly followed by a value. From the hundred interviews on beef and yogurt, and fifty interviews on olive oil and vegetable oil that were conducted, 128 ladders on beef, 148 ladders on yogurt, 60 ladders on olive oil and 47 ladders on vegetable oil, containing complete ACV-links, were retained. The number of different attributes, consequences and values after content analysis and categorization are depicted in Table 1. The ladders are used to construct a 3-way $A_i C_j V_k$ contingency table for each product. The cells of the 3-way table contain the frequencies with which each of the linkages occurred in the data.

Table 1
Characteristics of the laddering data used to test for conditional independence

	Beef	Yogurt	Olive oil	Vegetable oil
#interviews	100	100	50	50
Attributes	10	10	8	5
Consequences	11	12	5	5
Values	13	17	8	5

Since in the four contingency tables the sample size is less than 5 times the number of cells in the table, the condition of Larntz (1978) is violated. We therefore adjust the χ^2 test statistics according to Williams' formula (Williams, 1976). Table 2 gives normed fit indices Δ, correction factors, the adjusted χ^2-statistics for both the unconditional $[AC, CV, AV]$ and conditional $[AC, CV]$ model, as well as the adjusted χ^2-difference test-statistics for the tests of conditional independence [4]. The adjusted χ^2 values for both the $[AC, CV, AV]$ and the $[AC, CV]$ models are low, indicating that the models fit the data very well. This is supported by the normed fit indices being close to one, indicating that there is not much improvement in fit possible. In each of the four tests, the assumption of conditional independence cannot be rejected. For all products, the p-value is very close to one. This provides strong empirical evidence for the independence of both the AC- and CV-matrices and therefore supports the partitioning of the ladders as in APT. Our findings indicate that attributes and values are associated indirectly through the attribute–consequence and consequence–value linkages [5].

[4] Since the tables contain empty cells we have added a small constant (0.5 as recommended by Goodman (1970)) to each of the cells. In this way these cells are treated as sampling zeros (as opposed to structural zeros).

[5] The test of $[AC, CV]$ versus the saturated $[ACV]$ model is denoted as the unconditional test (Sobel, 1995), and provides an alternative to the test of conditional independence $[AC, CV]$ versus $[AC, CV, AV]$ presented by us. For all products, the unconditional test rejects the hypothesis of conditional dependence, supporting the findings of the conditional test.

Table 2
Tests for conditional independence of attributes and values given the consequences

	AC, CV, AV				*AC, CV*				*AC, CV* vs. *AC, CV, AV*			
	Δ	corr.	χ^2_{adj}	df	Δ	corr.	χ^2_{adj}	df	corr.	$\Delta\chi^2_{adj}$	df	*p*-value
Beef	0.957	3.41	19.47	1080	0.934	3.21	30.11	1188	1.20	25.15	108	> 0.999
Yogurt	0.932	3.90	28.18	1584	0.905	3.68	39.15	1728	1.22	27.83	144	> 0.999
Olive oil	0.950	2.35	8.16	196	0.905	2.13	15.41	245	1.23	11.08	49	> 0.999
Vegetable oil	0.962	1.77	4.35	64	0.919	1.64	9.42	80	1.13	6.88	16	0.975

5.2. Convergent validity

Since the independence assumption is supported, the models in Eqs. (2) and (3) can be estimated separately. In assessing the convergent validity of APT and laddering, we use all *AC*- and *CV*-linkages from the laddering interviews on beef and compare these data with linkages from the APT interviews on beef. We compare the number of entries in the *AC*- and *CV*-matrices in APT with the number of direct and indirect *AC*- and *CV*-linkages from the laddering implication matrix. For laddering, 462 links between attributes and consequences were observed, and 1068 links between consequences and values. In APT these figures were 5773 and 9670, respectively.

Both laddering and APT data were combined and organized in two 3-way contingency tables ($A_iC_jT_l$ and $C_jV_kT_l$) containing the frequencies of the A_iC_j- and C_jV_k-linkages respectively. In both contingency tables, the ratio of observations to number of cells exceeds 15. Hence, the condition suggested by Larntz (1978) is amply satisfied and the statistics do not need to be adjusted for cell sparseness as in Table 2. Tables 3 and 4 contain the test statistics for the models nested within Eqs. (2) and (3) and the statistical test results investigating the convergent validity of both approaches.

Table 3 reveals that the tests for similarity in content with respect to the attributes ([*A, C, T*] versus [*C, AT*]), consequences ([*A, C, T*] versus [*A, CT*]), and attributes and consequences simultaneously ([*A ,C, T*] versus [*AT, CT*]) are all highly significant ($p < 0.001$). This suggests that the content of the *AC*-part of the means-end chains differs significantly between the two measurement techniques.

Similarity of the structure is investigated by testing the model [*AC, AT, CT*] against the saturated model. The $\Delta\chi^2$-statistic is insignificant at the 5% level and the [*AC, AT, CT*] model describes the data sufficiently well ($\Delta = 0.98$). Hence, for the *AC*-part of the data, we conclude that similarity in structure is supported.

Table 4 shows the results of the model testing for the *CV*-part of the data. It shows the same picture as the *AC*-part. The three tests for the similarity of the content of the *CV*-part all reject the hypothesis at overwhelmingly low *p*-values. On the contrary, the $\Delta\chi^2$-statistic is insignificant when testing similarities of structure and the corresponding [*CV, CT, VT*] model fits the data quite well ($\Delta = 0.98$). Thus, similarity in structure is supported for the *CV*-part.

We should note here that, statistically speaking, the lack of fit of models used to test the similarity in

Table 3
Tests for similarity of APT and laddering, regarding content and structure of the *AC*-part of the means-end chains network

Model	Δ	χ^2	df	*p*-value	Test	$\Delta\chi^2$	df	*p*-value	Test for
[*A, C, T*]	0.807	1994.69	337	< 0.001					
[*C, AT*]	0.821	1844.56	330	< 0.001	[*A, C, T*] vs. [*C, AT*]	150.12	7	< 0.001	content
[*A, CT*]	0.824	1810.03	315	< 0.001	[*A, C, T*] vs. [*A, CT*]	184.65	22	< 0.001	content
[*AT, CT*]	0.839	1659.91	308	< 0.001	[*A, C, T*] vs. [*AT, CT*]	334.78	29	< 0.001	content
[*AC, AT, CT*]	0.982	180.49	154	0.071	[*AC, AT, CT*] vs. [*ACT*]	180.49	154	0.071	structure

Table 4
Tests for similarity of APT and laddering, regarding content and structure of the *CV*-part of the means-end chains network

Model	Δ	χ^2	df	*p*-value	Test	$\Delta\chi^2$	df	*p*-value	Test for
[*C*, *V*, *T*]	0.726	3660.78	643	< 0.001					
[*V*, *CT*]	0.771	3063.83	624	< 0.001	[*C*, *V*, *T*] vs. [*V*, *CT*]	596.95	19	< 0.001	content
[*C*, *VT*]	0.748	3376.41	627	< 0.001	[*C*, *V*, *T*] vs. [*C*, *VT*]	284.37	16	< 0.001	content
[*CT*, *VT*]	0.792	2779.46	608	< 0.001	[*C*, *V*, *T*] vs. [*CT*, *VT*]	881.32	35	< 0.001	content
[*CV*, *CT*, *VT*]	0.976	321.76	304	0.232	[*CV*, *CT*, *VT*] vs. [*CVT*]	321.75	304	0.232	structure

content ([*A*, *C*, *T*], [*A*, *CT*], [*C*, *AT*], [*AT*, *CT*], [*C*, *V*, *T*], [*V*, *CT*], [*C*, *VT*] and [*CT*, *VT*]) is significant. Therefore the chi-square difference test statistics for similarity in content do not follow a central chi-square distribution, are formally inappropriate and tend to reject the null-model too quickly. However, the extremely small *p*-values of the $\Delta\chi^2$ test statistics suggest that differences do indeed exist in the content of the cognitive structures as measured by laddering and APT. The normed fit indices for these models are very close to the 0.8-level as suggested by Zahn and Fein (1979) as a cut-off level for models being acceptable. In addition, as discussed in Section 3.2, there is strong empirical support for the differences in the frequencies with which the concepts occur.

6. Discussion and conclusions

In this study we described APT as a quantitative technique for measuring means-end chains, and provided evidence on the validity of APT. Two major issues were addressed: the assumption of conditional independence of attributes and values given the consequences, and the convergent validity of APT as compared to the more traditional laddering interview.

The findings indicate that, across four product categories, attributes and values are conditionally independent. This supports the APT approach in which *AC*- and *CV*-linkages are measured separately. Our finding is in line with the contention in means-end chain theory that the link between the consequences and the values concerns the self of the consumer and is independent of the product involved (Walker and Olson, 1991).

With respect to between-method convergent validity, the results show that APT and laddering differ in the content (frequency of occurrence of specific concepts) of the means-end chains network identified. In general, APT yielded higher frequencies of occurrence of concepts than laddering. This is expected, given that APT involves recognition, and laddering recall. It cannot be stated a priori which technique yields the most valid information on content. In a recognition task, people may indicate more concepts than are really relevant to them. On the other hand, in a recall task, people may overlook important concepts, construct concepts during the interview, or be responsive to interviewer bias. Therefore, it is particularly important and reassuring that the strengths of the linkages (structure), corrected for the frequency of specific concepts, do not differ significantly between laddering and APT.

In sum, this study provides support for the validity of APT to uncover means-end chains. Our findings do not suggest that laddering is not an important and useful technique for measuring means-end chains. Laddering was never intended to be used with large, representative samples. It is a qualitative technique with all its advantages and disadvantages. APT is geared towards being used in large-scale quantitative surveys in which representative samples are required. However, applying APT requires exploratory research to identify the main concepts to be included in the *AC*- and *CV*-matrices. Laddering is the primary technique for this purpose [6]. Using APT in conjunction with laddering is in spirit with other approaches in marketing research, such as composi-

[6] Additionally, concept elicitation techniques like Kelly's repertory grid may be used to elicit both attributes and consequences. The values may be obtained from a value inventory such as the List of Values (Kahle, 1983). Nevertheless laddering seems the most suitable technique to develop the APT matrices as it is directly based on the means-end chain theory.

tional perceptual mapping and conjoint analysis, where qualitative techniques are used to identify key constructs that are subsequently summarized and used in fixed-format in large-scale studies (Green and Srinivasan, 1978; Dillon et al., 1985).

The findings reported in this study indicate that APT is a viable quantitative technique for means-end chain research. This does not mean that APT is without its limitations. A criticism that may be levied against APT is the simplified representation of the means-end chains network by merely considering linkages between concepts at adjoining levels, i.e., the *AC*- and *CV*-linkages. This limitation may be alleviated by adding extra *AA*-, *CC*- and *VV*-matrices, containing the same concepts in the rows as in the columns. However, given similarity of the structures revealed with APT and laddering, the benefits of the additional information obtained with these matrices may be questioned, while in addition respondent burden increases substantially. Future research may shed more light on this issue.

Several other avenues for future research may be identified. First, another way to alleviate the potentially simplified *A–C–V* representation of APT is to extend the APT approach to the six-level hierarchy proposed by Peter and Olson (1987): concrete/abstract attributes, functional/psychosocial consequences, and instrumental/terminal values. This would increase the data collection burden for the respondent. Whether this is worth the effort is an issue for future research.

Second, in this study we analyzed the linkages between the attributes, consequences and values ignoring whether ladders are elicited from the same or different subjects. A multi-level extension to the loglinear models proposed in this paper could be used to account for multiple ladders expressed by a single respondent. Recently, Poulsen and Valette-Florence (1996) proposed a heterogeneous latent Markov model to analyze means-end chains. The methodology enables the investigation of heterogeneity of means-end chains and, in that respect, extends the loglinear analysis performed in this study. Further research may examine the heterogeneity of perceived strengths of linkages between concepts and assess the validity of APT in a disaggregate setting.

Third, future research may focus on the underlying factors responsible for the differences in the occurrence of the concepts, especially the effects of recognition versus recall and the extent of unwanted strategic processes that may underlie APT. Finally, it is necessary to replicate this study with different products and in other countries to confirm and generalize our findings.

Acknowledgements

We are grateful to our anonymous reviewers for their encouraging and helpful comments and to the Commission of the European Communities for research support (grant no. AIR2-CT94-1066).

References

Abbott, L., 1955. Quality and competition. Greenwood Press, Westwood.

Agresti, A., 1990. Categorical data analysis. Wiley, New York.

Alba, J.W., Hutchinson, J.W., 1987. Dimensions of consumer expertise. Journal of Consumer Research 13, 411–454.

Aurifeille, J.M., Valette-Florence, P., 1995. Determination of the dominant means-end chains: A constrained clustering approach. International Journal of Research in Marketing 12, 267–278.

Bettman, I., 1974. Toward a statistics for consumer decision net models. Journal of Consumer Research 1, 71–80.

Bonett, D.G., Bentler, P.M., 1983. Goodness-of-fit procedures for the evaluation and selection of log-linear models. Psychological Bulletin 93, 149–166.

Claeys, C., Swinnen, A., Vanden Abeele, P., 1995. Consumers' means-end chains for 'think' and 'feel' products. International Journal of Research in Marketing 12, 193–208.

DeSarbo, W.S., Hildebrand, D.K., 1980. A marketer's guide to log-linear models for qualitative data analysis. Journal of Marketing 44, 40–51.

Dillon, W.R., Frederick, D.G., Tangpanichdee, V., 1985. Decision issues in building perceptual product spaces with multi-attribute rating data. Journal of Consumer Research 12, 47–63.

Fienberg, S., 1980. The analysis of cross-classified categorical data. MIT Press, Cambridge.

Goodman, L.A., 1970. The multivariate analysis of qualitative data: Interactions among multiple classifications. Journal of the American Statistical Association 65, 226–256.

Green, P.E., Srinivasan, V., 1978. Conjoint analysis in consumer research: Issues and outlook. Journal of Consumer Research 5, 103–123.

Grunert, K.G., 1997. What's in a steak? A cross-cultural study on the quality perception of beef. Food Quality and Preference 8, 157–171.

Grunert, K.G., Grunert, S.C., 1995. Measuring subjective meaning

structures by the laddering method: Theoretical considerations and methodological problems. International Journal of Research in Marketing 12, 209–225.

Gutman, J., 1982. A means-end chain model based on consumer categorization processes. Journal of Marketing 46, 60–72.

Gutman, J., 1991. Exploring the nature of linkages between consequences and values. Journal of Business Research 22, 143–148.

Kahle, L.R., 1983. Social values and social change: Adaptation to life in America. Praeger, New York.

Kelly, G.A., 1955. The psychology of personal constructs. W.W. Norton and Co. Inc., New York.

Larntz, K., 1978. Small-sample comparisons of exact levels for chi-squared goodness-of-fit statistics. Journal of the American Statistical Association 73, 253–263.

Margolin, B.H., Light, R.J., 1974. An analysis of variance for categorical data, II. Small sample comparisons with chi square and other competitors. Journal of the American Statistical Association 69, 755–764.

Pieters, R., Baumgartner, H., Allen, D., 1995. A means-end chain approach to consumer goal structures. International Journal of Research in Marketing 12, 227–244.

Peter, J.P., Olson, J.C., 1987. Consumer behaviour: Marketing strategy perspectives. Richard Irwin, Homewood (IL).

Poulsen, C., Valette-Florence, P., 1996. Improvements in means-end research: A heterogeneous latent Markov approach. Paper presented at the Marketing Science Conference, Gainesville, FL.

Reynolds, T.J., Craddock, A.B., 1988. The application of the MECCAS model to the development and assessment of advertising strategy. Journal of Advertising Research 28, 43–54.

Reynolds, T.J., Gutman, J., 1984. Advertising is image management. Journal of Advertising Research 24, 27–37.

Reynolds, T.J., Gutman, J., 1988. Laddering theory, method, analysis, and interpretation. Journal of Advertising Research 28, 11–31.

Schank, R.C., 1982. Dynamic memory: A theory of reminding and learning in computers and people. Cambridge University Press, New York.

Singh, S.N., Rothschild, M.L., 1983. Recognition as a measure of learning from television commercials. Journal of Marketing Research 20, 235–248.

Sobel, M.E., 1995. The analysis of contingency tables. In: Arminger, G., Clogg, C.C., Sobel, M.E. (Eds.), Handbook of statistical modeling for the social and behavioral sciences, Plenum Press, New York, pp. 251–303.

Steenkamp, J.E.B.M., van Trijp, H.C.M., 1996. Task experience and validity in perceptual mapping: A comparison of two consumer-adaptive techniques. International Journal of Research in Marketing 13, 265–276.

Tolman, E.C., 1932. Purposive behaviour in animals and men. Century, New York.

Valette-Florence, P., Rapacchi, B., 1991. Improvements in means-end chain analysis using graph theory and correspondence analysis. Journal of Advertising Research 31, 30–45.

Vanden Abeele, P., 1992. A means-end study of dairy consumption motivation. Report for the European Commission, EC regulation 1000/90 – 43 ST.

Walker, B.A., Olson, J.C., 1991. Means-end chains: Connecting products with self. Journal of Business Research 22, 111–118.

Williams, D.A., 1976. Improved likelihood ratio tests for complete contingency tables. Biometrika 63, 33–37.

Young, S., Feigin, B., 1975. Using the benefit chain for improved strategy formulation. Journal of Marketing 39, 72–74.

Zahn, D.A., Fein, S.B., 1979. Large contingency tables with large cell frequencies: A model search algorithm and alternative measures of fit. Psychological Bulletin 86, 1189–1200.